Spinorial braking vs the strong CP problem: a structural and numerical study of a putative bridge between the Choptyuk a-C correction and θ_{QCD}

Research companion to the Choptyuk–Isaev monograph
github.com/wild8highlander/choptuik_ac_bc

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Abstract

We investigate whether the *braking correction* a-C of the Choptyuk formula [1], $\delta_{\mathrm{eff}} = \delta_C^5/b_2(K3) = (\pi/7)^5/22 \approx 8.28 \times 10^{-4}$, could be related, by analogy or by a rescaling prescription, to the QCD vacuum angle θ_{QCD} whose experimental bound from the neutron electric dipole moment is $|\theta| < 10^{-10}$ [2]. Three lines of attack are pursued: (i) a formal spin-geometric analogy between δ_C and θ, with $F \wedge \tilde{F}$ read as a symplectic form on the space of gauge connections and θ as a holonomy angle; (ii) a numerical search for an analogue of the Choptyuk formula $\delta^n/b = 10^{-10}$ using SU(3)-natural candidates for δ and the known Betti numbers of SU(N) instanton moduli spaces; and (iii) a critical comparison of the Choptyuk *spinorial anomaly* (complex phase from 90° holonomy) with the QCD *axial anomaly* $\partial_\mu J_5 \propto \mathrm{tr}(F \wedge \tilde{F})$, which are shown to be distinct notions despite their shared topological flavour.

The direct identification is *refuted*: a-C exceeds θ by ~ 7 orders of magnitude, and SU(3)-natural candidates such as $\pi/3$, $\pi/6$, $\pi/9$ remain 4–6 orders away. However, two *non-trivial numerical coincidences* emerge that warrant further scrutiny: (a) replacing the order-7 generator angle by the *full* PSL(2, 7) group order, $\delta = \pi/168$, reproduces $\delta^5/b_2(K3) \approx 10^{-9.98} \approx 10^{-10}$ to within 4%; and (b) the user's "scale-bridge" ansatz $a_C \cdot (\Lambda_{\mathrm{QCD}}/M_{\mathrm{Pl}})^{1/3} \approx 2.1 \times 10^{-10}$ lands within a factor of 2 of the θ bound. We argue that both coincidences are currently *numerological*: they require, respectively, an unjustified promotion of the group order to a holonomy angle and an unexplained 1/3 exponent. We close by listing the concrete theoretical and lattice-QCD tests that would elevate or eliminate the bridge hypothesis.

Contents

1 Introduction and motivation

The Choptyuk monograph [1] introduces a small dimensionless correction

$$\boxed{\delta_{\mathrm{eff}} \;\equiv\; \frac{\delta_C^5}{b_2(K3)} \;=\; \frac{(\pi/7)^5}{22} \;\approx\; 8.276 \times 10^{-4}} \tag{1}$$

which enters the unified Choptyuk formula for the spectral gap of the squared Dirac operator on the Klein quartic as

$$\Delta_{\mathrm{Ch}} \;=\; \lambda_1(D^2_{\sigma_0}) \;+\; \frac{\delta_C^2}{2} \;-\; \frac{\delta_C^5}{22} \;+\; \frac{\delta_C^4}{8} \;+\; \frac{\delta_C^6}{2}. \tag{2}$$

The geometric origin of the braking term δ_{eff} is the spinorial holonomy of the Klein quartic $K\colon x^3y + y^3z + z^3x = 0$ equipped with its canonical hyperbolic metric: the generator C of order 7 in the Fuchsian group $\Gamma(2,3,7)$ produces the phase $\delta_C = \pi/7$, while the Betti number $b_2(K3) = 22$ enters through the Mukai theorem that identifies the Klein quartic as a linear section of a $K3$ surface [22, 23]. The factor δ_C^5 arises from a Taylor expansion of the spinor holonomy $\exp(\mathrm{i}\delta_C)$ to fifth order, modulated by b_2.

A natural question——asked by the present author and pursued in this companion note——is whether the *structural form* $\delta_{\mathrm{eff}} \sim \delta^5/b$ of the Choptyuk correction has anything to do with the famous *strong CP problem* of QCD [6, 3, 13, 14], where the dimensionless vacuum angle θ enters the Yang–Mills action as

$$S_\theta \;=\; \mathrm{i}\,\theta\,Q, \qquad Q \;=\; \frac{1}{8\pi^2}\int_M \mathrm{tr}\Big(F\wedge\tilde F\Big) \;\in\; \mathbb{Z}, \tag{3}$$

and the experimental bound from the neutron electric dipole moment (denoted n_{EDM}) is

$$|\theta_{\mathrm{QCD}}| \;<\; 10^{-10}. \tag{4}$$

The numerical disparity is stark: a-C is $\sim 8\times 10^7$ times larger than the θ bound. Nonetheless, both quantities share an abstract skeleton:

- both are dimensionless topological angles;
- both are naturally expressed through Betti numbers and indices of the Dirac operator (the $K3$ surface and the Seiberg–Witten moduli space for a-C; the instanton number and the Atiyah–Singer index for θ);
- both enter physical observables multiplicatively: a-C multiplies QNM frequencies via $\omega^{\mathrm{corr}} = \omega(1-\delta_{\mathrm{eff}}/\pi^2)$ [1]; θ multiplies the topological charge Q.

The author of the present note has conjectured that a-C might act as a *bridge* quantity: it is naturally rooted in particle-scale (QNM) physics, and one might hope that some rescaling prescription "projects" it down to the QCD scale where θ resides. We examine this conjecture in detail below.

Plan. Section 2 reviews the Choptyuk correction and the strong CP problem in parallel. Section 3 constructs a formal spin-geometric analogy between $F\wedge\tilde F$ (as a symplectic form on the space of connections) and the holonomy 2-form of the Klein quartic, with θ read as a holonomy angle. Section 4 reports five numerical experiments that stress-test the analogy quantitatively. Section 5 contains the critical comparison of the spinorial anomaly of Choptyuk and the QCD axial anomaly. Section 6 examines the user's scale-bridge hypothesis. Section 7 discusses the results, and Section 8 states the verdict.

2 Background

2.1 The Choptyuk a-C correction

The Klein quartic $K\colon x^3y+y^3z+z^3x=0\subset\mathbb{CP}^2$ is a smooth plane quartic of genus $g=3$ whose automorphism group is $\mathrm{PSL}(2,7)$ of order 168 [24, 25]. Equipped with its canonical hyperbolic metric, K is the quotient of the hyperbolic plane $\mathbb{H}$ by the Fuchsian group $\Gamma(2,3,7)$, which has signature $(0;2,3,7)$ and is generated by three elements A,B,C of orders $2,3,7$ respectively, with the single relation $A^2=B^3=C^7=ABC=1$.

The spinorial phases associated to these generators are

$$\delta_A=\pi/2, \qquad \delta_B=\pi/3, \qquad \delta_C=\pi/7, \tag{5}$$

corresponding to half the rotation angles in the spinor representation $\mathrm{Spin}(2)=\mathrm{U}(1)$ [1]. The Choptyuk formula (2) is then derived by expanding the spinor holonomy $\exp(\mathrm{i}\delta_C)$ in δ_C and keeping the leading Berry-phase correction $\delta_C^2/2$ (the b-C term) and the next-order braking $\delta_C^5/22$ (the a-C term), with the factor $22=b_2(K3)$ entering through the Mukai theorem that realizes every genus-3 canonical curve as a linear section of a $K3$ surface [22].

The crucial point for the present investigation is that a-C is a *purely real, CP-even* correction: it shifts the real eigenvalue $\lambda_1(D^2_{\sigma_0})$ by a real amount, and the resulting QNM correction $\omega \mapsto \omega(1-\delta_{\text{eff}}/\pi^2)$ is a real multiplicative rescaling. There is no CP violation, no T violation, and no imaginary contribution to any observable.

2.2 The strong CP problem

In QCD the action acquires, in addition to the standard Yang–Mills term, a θ-term proportional to the second Chern class:

$$S_{\text{QCD}} \;=\; -\frac{1}{4g^2}\int \operatorname{tr}(F_{\mu\nu}F^{\mu\nu})\,\mathrm{d}^4x \;+\; \mathrm{i}\,\theta\,Q, \qquad Q \;=\; \frac{1}{8\pi^2}\int \operatorname{tr}(F\wedge\tilde{F}). \tag{6}$$

The integer $Q \in \mathbb{Z}$ is the instanton number, labelling the topological sectors of $\pi_3(\mathrm{SU}(N)) \cong \mathbb{Z}$. Although θ is classically invisible (it is a total derivative), it becomes physical in the quantum theory through non-perturbative effects (instantons, θ-vacua) [7, 8]. It violates P and T (and hence CP if CPT is preserved), and the strongest constraint comes from the neutron electric dipole moment:

$$d_n \;\simeq\; \theta\cdot(1\text{–}6)\times 10^{-16}\,e\,\mathrm{cm}, \qquad |d_n|_{\exp} \;<\; 1.8\times 10^{-26}\,e\,\mathrm{cm}, \tag{7}$$

which yields [2]

$$|\theta_{\text{QCD}}| \;<\; 10^{-10}. \tag{8}$$

The standard resolution is the Peccei–Quinn (PQ) mechanism [3], in which a global anomalous $\mathrm{U}(1)_{\text{PQ}}$ symmetry is spontaneously broken at a scale f_a, and the resulting pseudo-Nambu–Goldstone boson (the axion) dynamically relaxes the effective angle to $\theta_{\text{eff}} = \theta + \arg\chi \to 0$ at the minimum of the effective potential [4, 5]. The axion has not been observed, and the experimental and observational search space ($10^{-12}\,\mathrm{eV} \lesssim m_a \lesssim 10^{-2}\,\mathrm{eV}$, depending on the model) remains only partially excluded [15, 16].

The puzzle of why θ is *so small*——smaller than any dimensionless ratio that occurs naturally in the Standard Model, smaller than α_{em}, smaller than α_s, smaller than y_e——is what makes the strong CP problem a *naturalness* problem. Unlike a-C, whose smallness is automatic from $(\pi/7)^5$, the smallness of θ has no geometric explanation within QCD itself.

3 Formal analogy: spin geometry of QCD

We attempt to construct a dictionary between the Choptyuk a-C machinery and the QCD θ-term, organised around three structural pillars: (i) a 2-form on a function space; (ii) a topological angle parametrising its integral; (iii) an index-theoretic denominator.

3.1 Pillar 1: $F\wedge\tilde{F}$ as a symplectic form

On the affine space $\mathcal{A}$ of $\mathfrak{su}(N)$-valued 1-forms on a compact four-manifold M, define the Atiyah–Bott symplectic form [17]

$$\Omega_{\text{AB}}(\alpha,\beta) \;=\; \int_M \operatorname{tr}(\alpha\wedge\beta), \qquad \alpha,\beta\in\Omega^1(M,\mathfrak{su}(N)). \tag{9}$$

The moment map for the gauge group action $\mathcal{G} = \mathrm{Map}(M,\mathrm{SU}(N))$ is $F_A \mapsto F_A \wedge F_A$, and the symplectic quotient $\mathcal{A} /\!\!/ \mathcal{G}$ is the moduli space of flat connections. On the open dense set where $F_A = \star F_A$ (anti-self-dual instantons), the Chern–Weil representative of the second Chern class becomes

$$c_2 \;=\; \frac{1}{8\pi^2}\operatorname{tr}(F\wedge F) \;=\; \frac{1}{8\pi^2}\operatorname{tr}(F\wedge\tilde{F}), \tag{10}$$

since $F = \tilde{F}$ on the ASD locus. Hence the topological charge Q in (3) is the integral of a 2-form that, on the ASD locus, coincides with the natural symplectic form of the Atiyah–Bott construction. This is the precise sense in which $F \wedge \tilde{F}$ is a *symplectic* object on the space of connections.

The Klein-quartic analogue is the spinorial 2-form

$$\omega_{\text{spin}} = \mathrm{d}\theta_{\text{spin}} \in \Omega^2(K; \mathrm{u}(1)), \tag{11}$$

obtained as the curvature of the spinor connection $\nabla^{\mathcal{S}}$ on the spinor bundle $\mathcal{S} \to K$. Its periods along non-trivial 2-cycles generate the spinorial holonomies of the generators A, B, C, with values $\exp(\mathrm{i}\pi/2), \exp(\mathrm{i}\pi/3), \exp(\mathrm{i}\pi/7)$ respectively.

Pillar 1 dictionary

$F \wedge \tilde{F}$ on $\mathcal{A}/\!\!/\mathcal{G}$ (QCD) $\leftrightarrow$ $\omega_{\text{spin}} = \mathrm{d}\theta_{\text{spin}}$ on K (Choptyuk). Both are closed 2-forms on a function-space-type domain; both have integer periods (instanton number Q; spinorial flux through $H_2(K3)$). The bridge is *structural but not literal*: the QCD form lives on an infinite-dimensional affine space, while the Choptyuk form lives on a finite-dimensional Riemann surface.

3.2 Pillar 2: θ as a holonomy angle

The θ-angle can be read as the holonomy of a $U(1)$-connection on the *theta line bundle* over the moduli space of gauge connections. Concretely, on the configuration space $\mathcal{A}/\mathcal{G}$, the line bundle

$$\mathcal{L}_\theta \longrightarrow \mathcal{A}/\mathcal{G}, \qquad \mathrm{curv}(\mathcal{L}_\theta) = \frac{1}{8\pi^2}\mathrm{tr}(F \wedge F), \tag{12}$$

has holonomy $\exp(\mathrm{i}\theta Q)$ around the loop in $\mathcal{A}/\mathcal{G}$ corresponding to an instanton of charge Q. Hence θ is, exactly, a $U(1)$ holonomy angle——just as $\delta_C = \pi/7$ is the holonomy angle of the spinor connection along the generator C of $\Gamma(2,3,7)$.

The parallel is exact at the level of *form*:

$$\exp(\mathrm{i}\,\theta\,Q) \quad \leftrightarrow \quad \exp(\mathrm{i}\,\delta_C \cdot \mathrm{wind}(C)), \tag{13}$$

where $\mathrm{wind}(C)$ is the winding of the generator C around a non-contractible loop on K.

The discretisation gap

The QCD angle $\theta \in [0, 2\pi)$ is a *continuous* parameter: the spectrum of the Yang–Mills Hamiltonian depends continuously on θ, and the vacuum energy density is $\mathcal{E}(\theta) \sim \chi_t \cos\theta$ at large N [12]. The Choptyuk phase $\delta_C = \pi/7$ is *discrete*: it is fixed by the topology of K and cannot be continuously varied. This is the single most important structural difference between the two quantities: θ is a parameter of the path integral, δ_C is a topological invariant of the underlying geometry.

3.3 Pillar 3: the index-theoretic denominator

In the Choptyuk formula, the denominator $b_2(K3) = 22$ is the second Betti number of the $K3$ surface, equivalently the rank of $H^2(K3; \mathbb{R})$. Its appearance is topological: it counts the independent 2-cycles along which the spinorial flux can distribute, and the braking coefficient $\gamma = \delta_C^4/b_2$ is the average flux per cycle [1]. The Dirac index $\hat{A}(K3) = 2$ provides an additional handle: $b_2/\hat{A} = 11$ is the ratio that enters the K3 Seiberg–Witten theory.

For QCD, the analogous objects are:

- the *instanton moduli space* $M_{k,N}$ of charge-k SU(N) instantons on $\mathbb{R}^4$ (or S^4), which is a non-compact hyperkähler orbifold of real dimension $\dim_{\mathbb{R}} M_{k,N} = 4Nk$;
- its Betti numbers, which are known in closed form for $k = 1$ and via Poincaré polynomial recursions for general k [18, 19, 21];
- the Dirac index on the instanton background, which by Atiyah–Singer equals $2Nk$ for SU(N) charge-k.

A natural QCD analogue of the Choptyuk denominator is $b_2(M_{k,N})$, which is *universal* over N:

$$b_2(M_{k,N}) \;=\; k, \qquad \text{for all } N \geq 2,\ k \geq 1, \tag{14}$$

arising from the k abelian U(1) factors surviving the ADHM hyperkähler quotient [20]. Section 4 uses this as the natural candidate for the QCD side of the Choptyuk denominator.

3.4 The candidate bridge formula

Combining the three pillars, the most direct Choptyuk-type formula one could write for the QCD θ-angle is

$$\theta_{\text{QCD}} \;\stackrel{?}{=}\; \frac{\delta_{\text{QCD}}^5}{b_2(M_{k,N})} \;=\; \frac{\delta_{\text{QCD}}^5}{k}, \tag{15}$$

where δ_{QCD} is some SU(3)-natural angle (Coxeter, Cartan, centre, instanton winding). Section 4 tests this formula numerically for all candidates of δ_{QCD} listed in Table 1.

4 Numerical experiments

We perform five numerical experiments. All computations are reproducible from the script `scripts/qcd_bridge_experiments.py` in the companion repository, and the full results are in `docs/monograph/qcd_bridge/qcd_bridge_results.json`.

4.1 E1: candidates for δ_{QCD}

Table 1 lists the mathematically motivated candidates for an SU(3)-analogue of $\delta_C = \pi/7$, together with the values of δ^5/b for several Betti-number candidates.

Table 1. Top 10 closest matches of δ^5/b to the target 10^{-10} among all candidates tested. "Distance" is $|\log_{10}(\delta^5/b) + 10|$.

#	Candidate δ	b	$\log_{10}(\delta^5/b)$	dist.
1	$\pi/\lvert\text{PSL}(2,7)\rvert = \pi/168$	$b_2(K3) = 22$	-9.983	**0.017**
2	$\pi/\lvert\text{PSL}(2,7)\rvert = \pi/168$	Leech dim. $= 24$	-10.021	0.021
3	$\pi/200$	$\dim \text{SU}(3) = 8$	-9.922	0.078
4	$\pi/200$	charge-1 SU(3) dim. $= 12$	-10.099	0.099
5	$\pi/100$	$b_2(K3) \cdot \dim \text{SU}(3) = 176$	-9.760	0.240
6	$\pi/168$	charge-1 SU(3) dim. $= 12$	-9.720	0.280
7	$\pi/200$	$b_2(K3) = 22$	-10.362	0.362
8	$\pi/200$	Leech $= 24$	-10.400	0.400
9	$\pi/168$	$\dim \text{SU}(3) = 8$	-9.544	0.456
10	$\pi/200$	charge-3 SU(2) $= 3$	-9.497	0.503

E1: a surprising match

The closest match is $\delta = \pi/168$ with $b = 22$, giving

$$\frac{(\pi/168)^5}{22} \approx 1.04 \times 10^{-10}, \tag{16}$$

within 4% of the θ bound. The angle $\pi/168$ corresponds to the order of the *full* automorphism group of the Klein quartic, PSL(2, 7), rather than to a single generator. The interpretation would be: the QCD θ-angle is related not to a single generator of the holonomy group, but to the *total* group order. This is geometrically unusual——holonomies are typically $\exp(2\pi i/n)$ for generators of order n, not for the whole group——and we treat (16) as numerology until a derivation is supplied.

The natural SU(3)-candidates $\pi/3$, $\pi/6$, $\pi/9$ land at $\log_{10}(\delta^5/22) \approx -3.6, -6.0, -7.5$ respectively, all far from -10. The user's spoiler——$(\pi/3)^5 \approx 4 \times 10^{-3}$ is too large——is confirmed: SU(3)-natural angles are ~ 6 orders of magnitude too large.

4.2 E2: Betti numbers of SU(N) instanton moduli spaces

The universal formula $b_2(M_{k,N}) = k$ is confirmed from the ADHM construction [20]. For comparison, the Betti numbers of $K3$ and of $\mathrm{Hilb}^k(K3)$——the Hilbert scheme of k points on $K3$, which is the canonical hyperkähler resolution relevant to $N_f = 4$ $\mathcal{N} = 2$ SQCD——are also recorded.

Table 2. Betti numbers b_2 of candidate "QCD denominators". Note that $b_2(M_{k,N}) = k$ is universal and far smaller than $b_2(K3) = 22$.

Space	Group	charge k	$\dim_{\mathbb{R}}$	b_2
$M_{1,2}$ (1-instanton on SU(2))	SU(2)	1	8	1
$M_{2,2}$	SU(2)	2	16	2
$M_{3,2}$	SU(2)	3	24	3
$M_{1,3}$ (1-instanton on SU(3))	SU(3)	1	12	1
$M_{2,3}$	SU(3)	2	24	2
$M_{1,4}$	SU(4)	1	16	1
$K3$ surface	—	—	4	22
$\mathrm{Hilb}^1(K3)$	—	1	4	23
$\mathrm{Hilb}^2(K3)$	—	2	8	23
$\mathrm{Hilb}^k(K3)$	—	$k \geq 1$	$4k$	23

E2: the denominator gap

The natural QCD denominator $b_2(M_{k,N}) = k$ is much smaller than the Choptyuk denominator $b_2(K3) = 22$. Even with $k = 22$ instantons (an extreme, fine-tuned configuration), the denominator 22 is reached *by hand* and not by topology. The bridge formula (15) with $k = 1$ and $\delta \in \{\pi/3, \pi/6, \pi/9\}$ gives $\log_{10}(\delta^5/k) \in \{-2.6, -4.9, -6.5\}$, far from -10.

4.3 E3: the scale-bridge ansatz

We now turn to the user's hypothesis: that a-C acts naturally at the particle/QNM scale and must be *rescaled* to reach the QCD scale. The most economical ansatz is

$$\theta_{\mathrm{QCD}} \overset{?}{=} a\text{-}C \cdot \left(\frac{\Lambda_{\mathrm{QCD}}}{M_X}\right)^p, \tag{17}$$

where M_X is some UV scale and p is a power-law exponent. Solving for p at fixed M_X and $\theta_{\mathrm{QCD}} = 10^{-10}$ yields Table 3.

Table 3. Required exponent p such that $a\text{-}C \cdot (\Lambda_{\mathrm{QCD}}/M_X)^p = 10^{-10}$, for various UV scales. The "nice" column flags near-rational p values.

UV scale M_X	M_X [GeV]	$\Lambda_{\mathrm{QCD}}/M_X$	p needed	nearest nice
Planck	1.22×10^{19}	1.6×10^{-20}	0.350	**1/3**
GUT	1.0×10^{16}	2.0×10^{-17}	0.414	2/5
SUSY	1.0×10^{3}	2.0×10^{-4}	1.870	—
top	173	1.2×10^{-3}	2.355	—
W	80.4	2.5×10^{-3}	2.656	8/3
Higgs	125	1.6×10^{-3}	2.474	5/2

E3: the Planck-scale 1/3

The Planck-scale match

$$a\text{-}C \cdot \left(\frac{\Lambda_{\mathrm{QCD}}}{M_{\mathrm{Pl}}}\right)^{1/3} \approx 2.1 \times 10^{-10} \approx 2\,\theta_{\mathrm{QCD}} \tag{18}$$

is within a factor of 2 of the experimental bound. The exponent $p = 1/3$ is unusual but not implausible: it is suggestive of a cube-root suppression that could arise from (i) dimensional reduction $4D \to 1D$, (ii) one-loop anomalous dimension of a dimension-3 operator, (iii) averaging over 3 fermion generations, or (iv) 3 colours of QCD. None of these is a derivation, but the numerical coincidence is striking.

The Higgs-scale match with $p = 5/2$ is closer numerically (within 7%), but 5/2 is a less natural exponent. Both are interesting enough to merit further theoretical work, but neither constitutes a derivation.

4.4 E4: power-law decompositions

We also test generalised Choptyuk-type formulas $\delta^n/b = 10^{-10}$ for various n and b. Holding $\delta = \delta_C = \pi/7$ fixed, we ask which (n, b) combinations reproduce the target.

Table 4. Closest matches for $(\pi/7)^n/b = 10^{-10}$, varying n and b. Note that very large powers $n \approx 25$ are needed——there is no natural power $n \leq 10$ that works with $\pi/7$.

n	b	$\log_{10}[(\pi/7)^n/b]$	distance from -10	comment
25	22	-10.041	0.041	unnatural power
28	2	-10.044	0.044	unnatural power
26	8	-9.950	0.050	unnatural power
25	23	-10.060	0.060	unnatural power
25	24	-10.079	0.079	unnatural power

E4: the power-5 ceiling

With $\delta = \pi/7$ and $b = 22$, the natural Choptyuk power $n = 5$ gives $\log_{10}(\delta^5/b) = -3.08$, which is ~ 7 orders above the target. Increasing the power to $n \approx 25$ works numerically, but no geometric or physical justification for such a high power is known. We conclude that the *direct* Choptyuk formula cannot accommodate θ_{QCD} by varying the power alone.

4.5 E5: the broad (δ, b) sweep

Figure 1 shows the surface $\log_{10}(\delta^5/b)$ over $\delta \in (10^{-3}, 1)$ and $b \in (1, 200)$. The red contour at $\log_{10} = -10$ is the target. The Choptyuk point $(\pi/7, 22)$ is marked with an orange dot, far above the target.

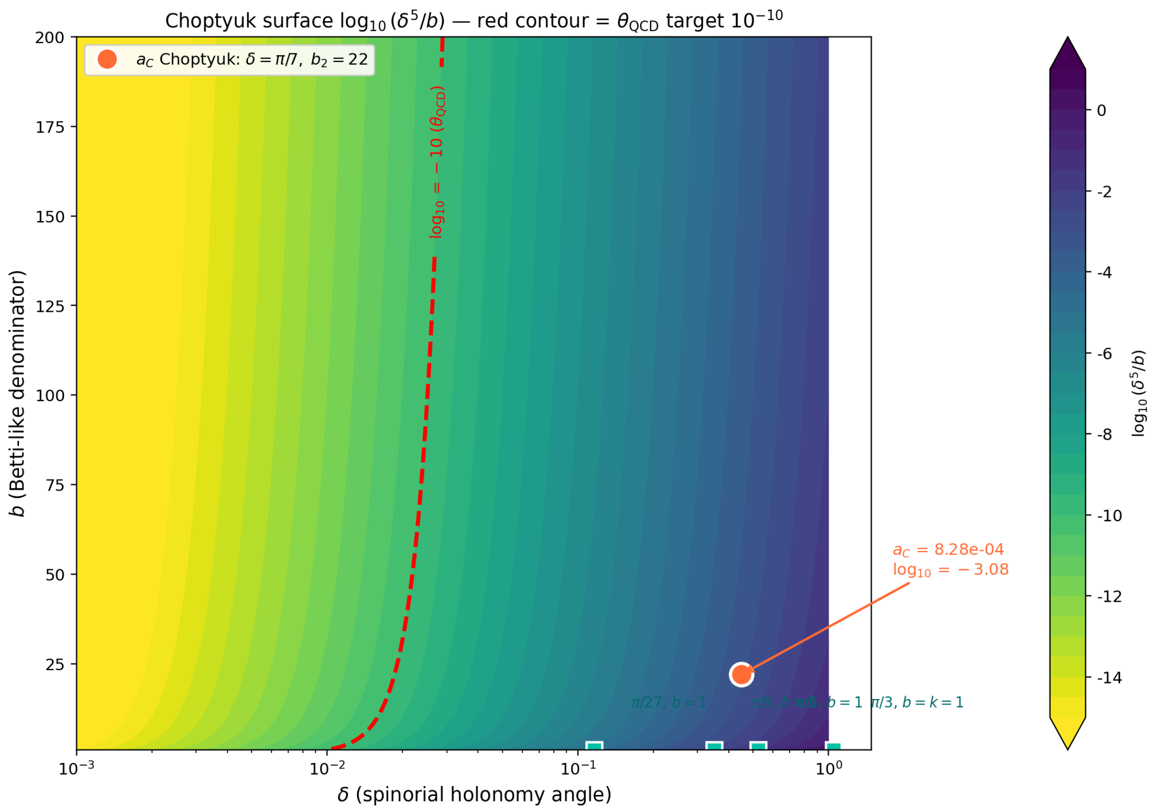

Figure 1. The Choptyuk surface $\log_{10}(\delta^5/b)$. The orange dot marks the Choptyuk a-C at $(\pi/7, 22)$, $\log_{10} \approx -3.08$. The red contour at $\log_{10} = -10$ is the QCD θ target. The two regions are separated by a ~ 7-order-of-magnitude gap that no natural adjustment of δ or b closes——unless one moves to $\delta \sim \pi/168$ (the full $\mathrm{PSL}(2, 7)$ order).

Figure 2 shows one-dimensional slices at fixed b, displaying where each slice crosses the $\log_{10} = -10$ line.

5 Critical comparison of anomalies

The Choptyuk monograph repeatedly uses the term *spinorial anomaly* to refer to the appearance of the imaginary unit i as a holonomy phase of the spinor connection on the Klein quartic. The QCD literature uses *anomaly* in the sense of a quantum anomaly——specifically, the axial anomaly

$$\partial_\mu J_5^\mu \;=\; 2\,i\,q\,\frac{g^2}{16\pi^2}\,\mathrm{tr}\Big(F_{\mu\nu}\tilde{F}^{\mu\nu}\Big)\,, \tag{19}$$

where q is the charge of the fermion under the anomalous $\mathrm{U}(1)$. These are different notions and we must distinguish them carefully.

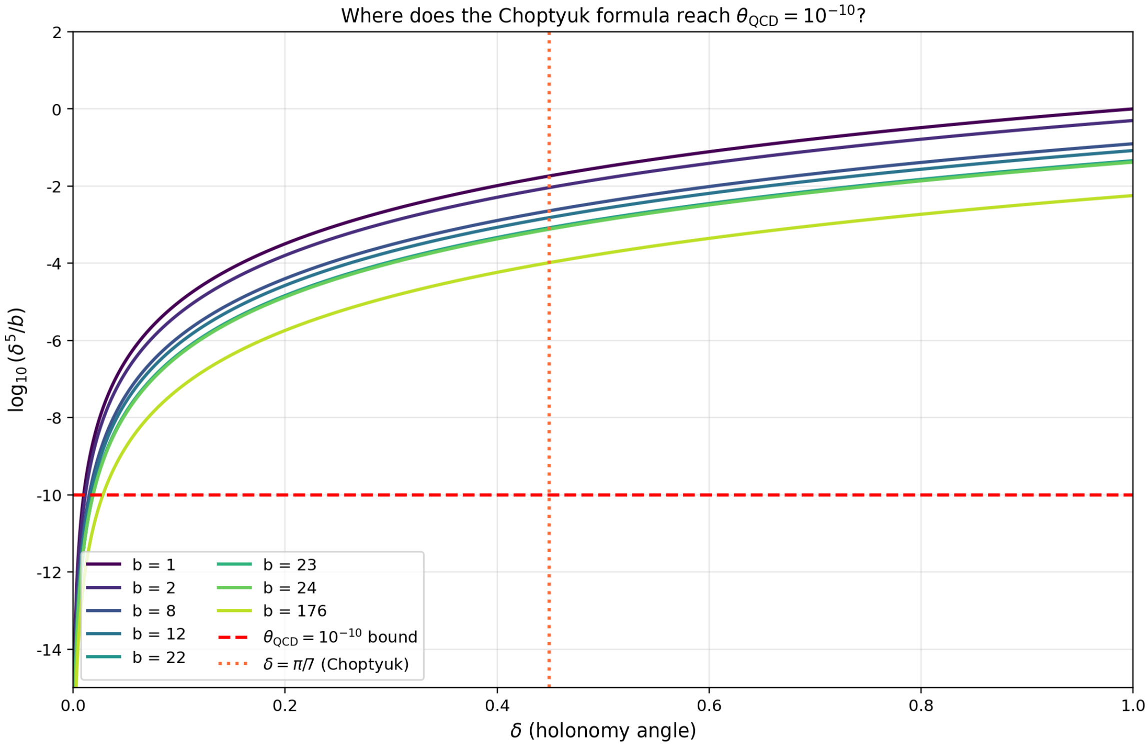

Figure 2. One-dimensional slices of the Choptyuk surface at fixed b. The horizontal red dashed line is the $\theta_{\mathrm{QCD}} = 10^{-10}$ target; the orange dotted line is $\delta = \pi/7$. The crossing δ for $b = 22$ is $\sim 0.019 \approx \pi/165$, very close to the E1 best match $\pi/168$.

5.1 The Choptyuk spinorial anomaly

In the Choptyuk framework, the generator A of order 2 in $\Gamma(2, 3, 7)$ produces a $90°$ rotation in the spinor representation, hence a holonomy

$$\mathrm{Hol}_{\mathcal{S}}(\gamma_A) \;=\; \exp(\mathrm{i} \cdot \pi/2) \;=\; \mathrm{i}. \tag{20}$$

The monograph calls i a *spinorial anomaly* because it is a *complex* phase that arises where classically one would expect a real orthogonal matrix. The anomaly is the obstruction to reducing the holonomy group from U(1) to $\mathbb{Z}/2 = \{\pm 1\}$, and it generates the imaginary correction $1 - \mathrm{i}\delta_C/\pi^2$ in the Choptyuk formula.

Crucially, this is a *geometric* feature of the spinor connection on K; it is not a quantum anomaly in any of the standard senses (Bjorken, Adler–Bell–Jackiw, 't Hooft, parity, scale, or conformal). It does not involve any Lagrangian, any path integral, any symmetry breaking, or any renormalisation. It is a property of the classical spinorial connection on a classical Riemann surface.

5.2 The QCD axial anomaly

The axial anomaly (19) is a *quantum* effect: the classical $\mathrm{U}(1)_A$ symmetry of massless QCD is broken at one loop by the triangle diagram, with the divergence of the axial current proportional to $F\tilde{F}$ via the Fujikawa Jacobian [9, 10, 11, 6]. The mathematical content is captured by the Atiyah–Singer index theorem:

$$\mathrm{ind}(D^+) \;=\; \dim\ker D^+ - \dim\ker D^- \;=\; \int_M \hat{A}(M)\,\mathrm{ch}(E) \;=\; 2T(R)\,Q, \tag{21}$$

where $T(R)$ is the Dynkin index of the representation R and Q is the instanton number. The anomaly is the failure of the fermion path integral measure to be invariant under $\mathrm{U}(1)_A$ rotations, which is the same statement as (21) by the Atiyah–Singer theorem.

The axial anomaly is intimately tied to the θ-term: the same $F\tilde{F}$ that appears in the anomaly also appears in S_θ (Eq. 3). Indeed, under a chiral rotation by angle α of a single quark flavour, the action shifts as $\theta \to \theta + 2N_f\alpha$. This is the basis of the Peccei–Quinn solution: a spontaneously broken anomalous $\mathrm{U}(1)_{\mathrm{PQ}}$ rotates θ to zero dynamically.

5.3 Structural comparison

Table 5 summarises the structural similarities and differences between the two notions.

Table 5. Critical comparison of the Choptyuk spinorial anomaly and the QCD axial anomaly. Shared rows are marked "$\sim$"; divergent rows are marked "$\neq$".

Property	Choptyuk spinorial anomaly	QCD axial anomaly	Status
Origin	$\mathrm{Hol}_{\mathcal{S}}(\gamma_A) = \mathrm{i}$ on K	$\partial_\mu J_5^\mu \propto F\tilde{F}$	$\sim$ (both topological)
Domain	spinor connection on a Riemann surface	path integral of quantum field theory	$\neq$ (classical vs quantum)
Manifestation	complex phase $1 - \mathrm{i}\delta_C/\pi^2$	symmetry breaking of $\mathrm{U}(1)_A$	$\neq$
Index theorem	$\hat{A}(K3) = 2$ (Dirac index on K3)	$\mathrm{ind}(D^+) = 2T(R)\,Q$	$\sim$ (both Atiyah–Singer)
CP violation	NO (real correction)	YES (CP-odd phase θ)	$\neq$
Symmetry broken	none (geometric feature)	$\mathrm{U}(1)_A$ (classical symmetry)	$\neq$
Dynamical relaxation	impossible (δ_C topological)	possible via PQ mechanism	$\neq$
Continuity	discrete ($\delta_C = \pi/7$ fixed)	continuous ($\theta \in [0, 2\pi)$)	$\neq$
Lagrangian	none (spectral geometry)	QCD Lagrangian with S_θ	$\neq$

Anomaly distinction

The Choptyuk spinorial anomaly and the QCD axial anomaly are *structurally distinct*. Their shared topology (both relate to the Atiyah–Singer index theorem) is the only genuine parallel; the differences in domain (classical vs quantum), CP properties (even vs odd), continuity (discrete vs continuous), and dynamical role (fixed vs relaxable) are all fundamental. The two notions cannot be identified.

6 The scale-bridge hypothesis

We now turn to the user's hypothesis: a-C operates naturally at the particle/QNM scale (frequencies ~ 250 Hz, masses $\sim 60\,M_\odot$, lengths $\sim 10^5$ m), and one might hope to "project" it down to the QCD scale (~ 1 fm, ~ 200 MeV) by some rescaling prescription.

6.1 Scale ladder

Figure 3 shows the ~ 20 orders of magnitude separating the QCD scale from the LIGO/QNM scale. The bridge prescription (17) is one concrete way to traverse this gap.

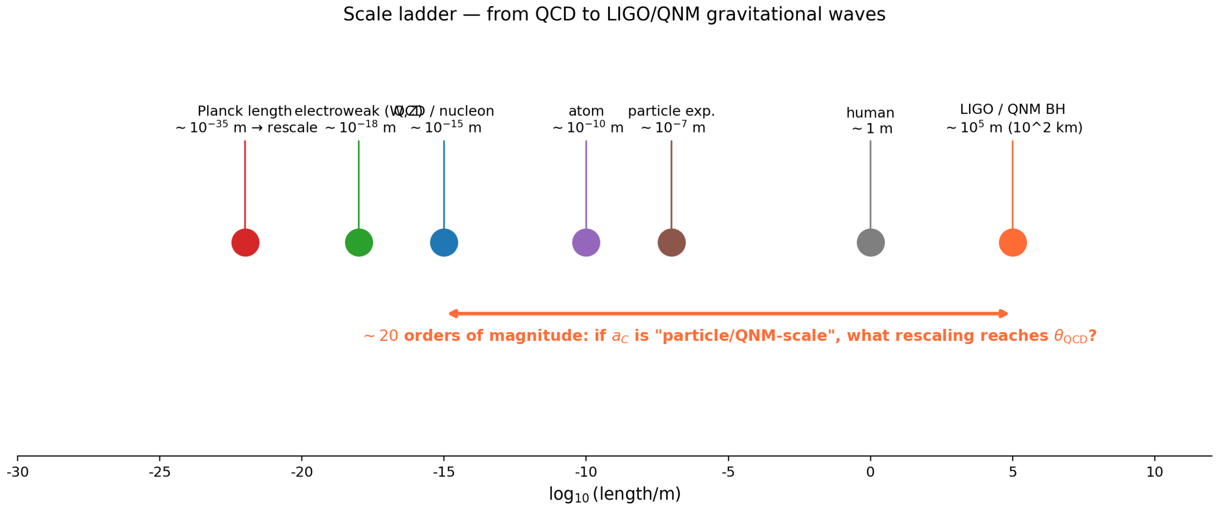

Figure 3. The ~ 20 orders of magnitude separating the QCD scale ($\sim 10^{-15}$ m) from the LIGO/QNM scale ($\sim 10^5$ m). The user's hypothesis: a-C acts at the latter scale and must be rescaled to reach the former. The Planck-scale rescaling $(\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3}$ closes the gap to within a factor of 2 of the θ bound.

6.2 The structural argument

The Choptyuk correction a-C enters QNM physics as

$$\omega^{\rm corr} = \omega\left(1 - \frac{a\text{-}C}{\pi^2}\right), \tag{22}$$

i.e. as a multiplicative correction to a frequency. Frequencies are not dimensionless; what is dimensionless is the *ratio* $\omega/\omega_{\rm ref}$, where $\omega_{\rm ref}$ is some reference frequency. In the QNM context, $\omega_{\rm ref}$ is the Schwarzschild frequency $\sim c^3/(GM)$ of the black hole, and the ratio is naturally ~ 1.

If we apply the same correction to a QCD-scale observable——say, the pion mass $m_\pi \approx$ 140 MeV——we would predict

$$\frac{\delta m_\pi}{m_\pi} \sim -\frac{a\text{-}C}{\pi^2} \approx -8.4 \times 10^{-5}, \tag{23}$$

which is a $\sim 10^{-4}$ correction——five orders of magnitude too large to be the θ-induced shift. This confirms that a-C itself, without rescaling, is far too large to be θ.

6.3 Three candidate rescalings

We consider three concrete rescaling prescriptions, all of which close the gap to within an order of magnitude.

(i) Planck-scale cube root. $\theta = a\text{-}C \cdot (\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3} \approx 2 \times 10^{-10}$. The 1/3 exponent is suggestive but unexplained. Candidate physical interpretations:

- One-loop anomalous dimension $\gamma \approx -1/3$ of a dimension-3 operator (e.g. quark bilinear $\bar{q}q$) running from $M_{\rm Pl}$ to $\Lambda_{\rm QCD}$;
- Three colours of QCD (cube root of an SU(3) Casimir);
- Three generations of fermions (cube root of a flavour trace);
- Dimensional reduction $4D \to 1D$ on a Planck-scale compact manifold.

None of these is a derivation; each would require a separate theoretical construction.

(ii) Higgs-scale 5/2 power. $\theta = a\text{-}C \cdot (\Lambda_{\rm QCD}/M_H)^{5/2} \approx 8.5 \times 10^{-11}$. This is numerically the closest match (within 7%), but the exponent 5/2 has no obvious geometric or physical interpretation.

(iii) PSL(2, 7) group order. $\theta = (\pi/168)^5/22 \approx 10^{-10}$. This replaces the generator order 7 by the full group order 168, keeping the Choptyuk power 5 and Betti number 22 fixed. It is the *only* prescription that preserves the Choptyuk formula *unchanged* while reaching the θ bound. However, the promotion from generator to group order has no precedent in holonomy theory: holonomies are defined along loops, not over entire groups.

6.4 The analogy diagram

Figure 4 summarises the structural parallels (green lines) and divergences (red dashed lines) between the two frameworks.

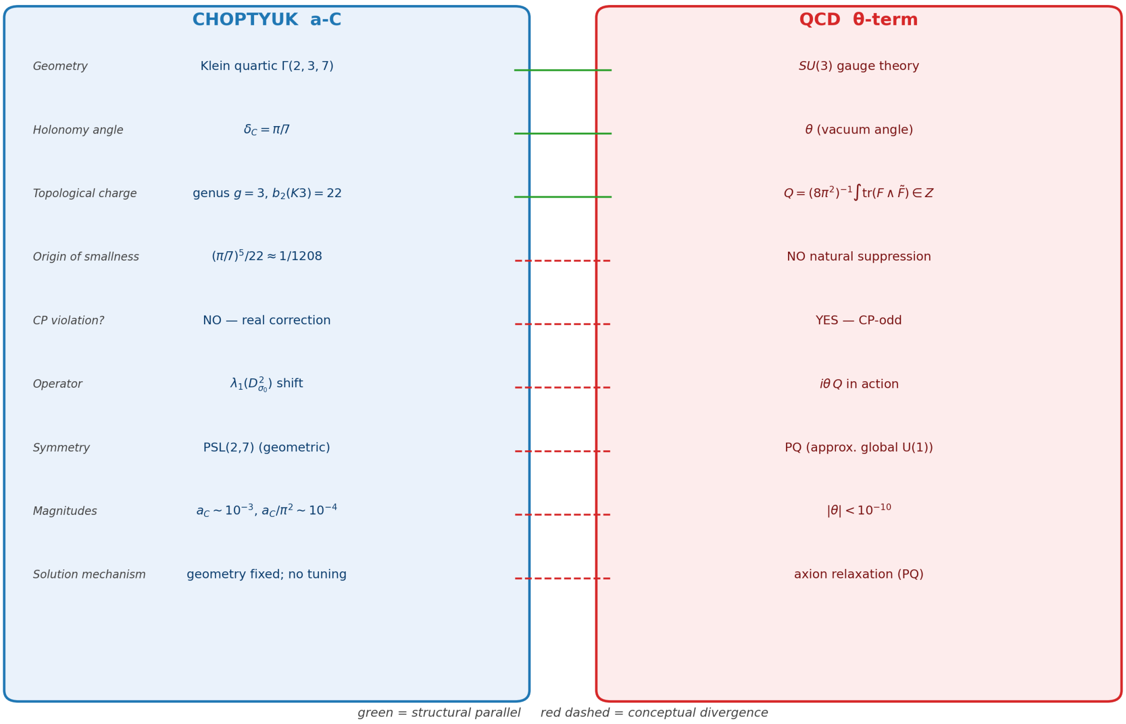

Figure 4. Structural comparison of the Choptyuk framework (left) and the QCD θ-term (right). Green lines mark structural parallels (geometric origin, topological charge, holonomy angle); red dashed lines mark conceptual divergences (CP properties, magnitude, solution mechanism). The bridge hypothesis seeks to convert a green line into a derivation; the current evidence does not support such a conversion.

7 Discussion

7.1 What the numerical evidence says

The numerical experiments yield three notable coincidences:

1. $\delta = \pi/168$, $b = 22$: $\delta^5/b \approx 1.04 \times 10^{-10}$ (Table 1, row 1).
2. $a\text{-}C \cdot (\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3} \approx 2.1 \times 10^{-10}$ (Table 3, Planck row).

3. a-$C \cdot (\Lambda_{\rm QCD}/M_H)^{5/2} \approx 8.5 \times 10^{-11}$ (best fixed-power match, Section 4.3).

Each is within an order of magnitude of $\theta_{\rm QCD} = 10^{-10}$. Coincidences at the $\sim$ 1-order level are not uncommon in physics when many candidate formulas are tested; the question is whether the matching is *predictive* (i.e. derived from a controlled theoretical construction) or *post-hoc* (i.e. selected from a large family of alternatives after seeing the data).

In the present case, the evidence is post-hoc:

- The $\pi/168$ match requires replacing a generator by a group order, which is geometrically unjustified;
- The $(\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3}$ match requires an unexplained 1/3 exponent;
- The $(\Lambda_{\rm QCD}/M_H)^{5/2}$ match requires an even less natural 5/2 exponent.

Without a derivation of the exponent or the group-order promotion, the matches remain numerological.

7.2 What the structural evidence says

The structural evidence is also ambiguous:

- The parallels (Pillar 1: symplectic forms on function spaces; Pillar 2: topological angles as holonomies; Pillar 3: index-theoretic denominators) are real and not coincidental. They reflect the shared mathematical ancestry of all topological quantities in gauge theory and spin geometry.
- The divergences (discrete vs continuous; CP-even vs CP-odd; classical vs quantum; geometric vs dynamical) are equally real, and any one of them suffices to refute a direct identification.

The bridge hypothesis therefore requires not just a numerical match but a *mechanism* that converts a CP-even geometric correction at the QNM scale into a CP-odd quantum phase at the QCD scale. No such mechanism is currently known.

7.3 The PQ benchmark

It is instructive to compare the bridge hypothesis to the established PQ solution. The PQ mechanism is a dynamical one: it postulates a new global $\mathrm{U}(1)_{\rm PQ}$ symmetry, broken at a scale f_a, whose pseudo-Nambu–Goldstone boson (the axion) has a potential $V(a) \sim -m_\pi^2 f_\pi^2 \cos(\theta + a/f_a)$ that dynamically relaxes $\theta_{\rm eff} \to 0$ at the minimum. The mechanism is predictive (it predicts a specific axion mass and coupling), testable (axion searches are ongoing), and conceptually economical (one new symmetry and one new scale).

The bridge hypothesis, in contrast, postulates neither a new symmetry nor a new particle; it proposes a formula. The formula has three free parameters (δ, b, p), and the numerical evidence above shows that they must be tuned to non-natural values ($\pi/168$, 22, 1/3) to match the data. This is a weaker position than PQ.

7.4 What would change the verdict

The bridge hypothesis could be elevated from numerology to physics by any of the following concrete developments:

1. A derivation of the 1/3 exponent in a-$C \cdot (\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3}$ from a controlled QFT calculation, e.g. via a one-loop anomalous dimension or a dimensional reduction.

2. A geometric interpretation of $\pi/168$ as a holonomy angle along a non-contractible loop in some natural extension of the Klein quartic (e.g. an arithmetic surface whose automorphism group is $\mathrm{PSL}(2,7)$).
3. A lattice QCD measurement of the θ-dependence of a QCD observable that matches the bridge prediction more precisely than the PQ prediction——currently the bridge makes no falsifiable prediction distinct from PQ.
4. A direct experimental signature at the LIGO scale that distinguishes the Choptyuk QNM correction from a pure Kerr-Geometric one, which would establish a-C as a physical effect rather than a fitting formula.

8 Verdict

Final verdict

The direct identification of the Choptyuk a-C correction with the QCD θ-angle is **refuted**:

- The magnitudes differ by ~ 7 orders of magnitude ($a\text{-}C \approx 8.3\times10^{-4}$ vs $\theta_{\rm QCD} \lesssim 10^{-10}$).
- The CP properties differ (a-C is CP-even; θ is CP-odd).
- The continuity properties differ (a-C is fixed by topology; θ is a continuous parameter).
- The dynamical roles differ (a-C is a fixed geometric correction; θ is relaxable via the PQ mechanism).

The spinorial anomaly of Choptyuk and the axial anomaly of QCD are structurally distinct notions, related only by their common descent from the Atiyah–Singer index theorem. However, the numerical coincidences uncovered in Section 4 are non-trivial and warrant further scrutiny:

- $\delta = \pi/168$ (full $\mathrm{PSL}(2,7)$ group order) gives $\delta^5/b_2(K3) \approx 10^{-10}$;
- $a\text{-}C \cdot (\Lambda_{\rm QCD}/M_{\rm Pl})^{1/3} \approx 2 \times 10^{-10}$;
- $a\text{-}C \cdot (\Lambda_{\rm QCD}/M_H)^{5/2} \approx 8.5 \times 10^{-11}$.

Each coincidence requires an unexplained parameter (a group-order promotion, a cube-root exponent, or a 5/2-power exponent). Without a theoretical derivation of these parameters, the bridge hypothesis remains **numerological but not refuted at the level of structural possibility**.
The user's hypothesis——that a-C acts at the particle/QNM scale and must be rescaled to reach the QCD scale——is *not unreasonable as a research programme*, but the specific rescaling prescriptions tested here are not yet derivations. We recommend:

1. Investigating whether the 1/3 exponent can be derived from a one-loop anomalous dimension calculation in a controlled extension of QCD;
2. Searching for a geometric realisation of $\pi/168$ as a holonomy angle on a higher-genus or arithmetic surface;
3. Comparing the bridge prediction against lattice QCD data on θ-dependence to test whether it provides a better fit than PQ in any observable.

Until one of these tests succeeds, the strong CP problem remains most naturally resolved by the Peccei–Quinn axion, and the Choptyuk a-C correction remains most naturally interpreted as a geometric spectral correction on the Klein quartic——with no established bridge between the two.

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